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When the Phones Go Home Predicting Verizon's Phone Financing Security

The Hive: Yining Mao | Jack Shi | Qixin Cui | Alex Bevan | Yizhuo Li

Agenda

- Business Problem
- Exploratory Data Analysis
- Our Modeling Approach and Choice
- Model Interpretation
- Business Case
- Summary & Timeline

Business Problem

One of Verizon's main revenue streams, phone financing, loses money when borrowers default on payments. Our objective is to investigate different models and identify a model that predicts phone default that maximizes the profit by calculating the business value and provide applicable suggestions.



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Exploratory Data Analysis

FICO and Down Payment Show the Strongest Relationships to Default Risk

Fig 1: Correlation Heatmap: Numeric Variables vs Default



- Fig 1 shows that customers who pay with **cash** tend to have **lower FICO scores** and **smaller down payments**, both of which are linked to a higher chance of default.
- Credit** customers usually have **higher credit scores** and **fewer delinquencies**, showing that payment behavior and credit-standing are key indicators of risk.

Temporal Artificial Signatures Indicate Data Simulation

Fig 2: Applications Over Time

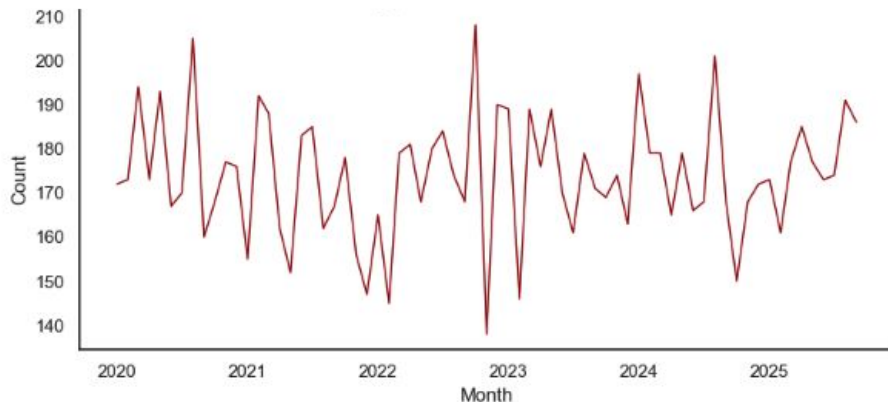


Fig 2 Fails to Reflect Real-World Seasonal Trends:

- Application volume is flat from 2020–2025 (showing no holiday spikes or new phone launch surges).
- No dips appear during events like COVID or economic slowdowns.
- Real-world sales usually rise and fall with market cycles and promotions.
- Missing these patterns limits how well the model captures true customer behavior.

Lack of Missing Data Also Indicates Simulation, Affecting Model Realism

Fig 3: Missing Data vs Not Missing by Field (% of Records)

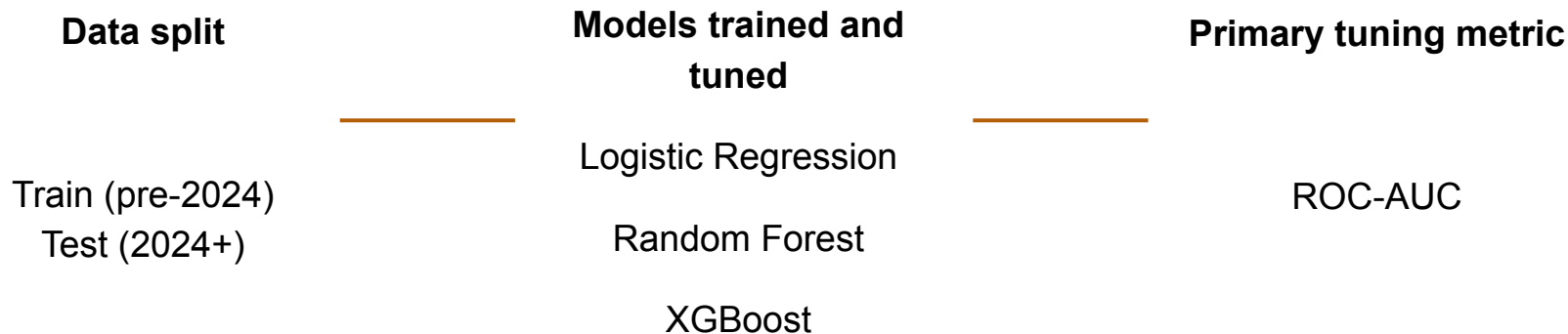


Fig 3 Shows No Missing Data (Artificial Completeness)

- Every variable has 0% missing values (unrealistic for real financial records)
- Real customer data usually includes gaps in income, FICO, or payment history
- Perfect completeness suggests the dataset was simulated
- Model accuracy might be distorted
- The data received seem to be simulated, with real Verizon data will show more natural variation. Our model can easily adapt and improve once those real-world patterns are included

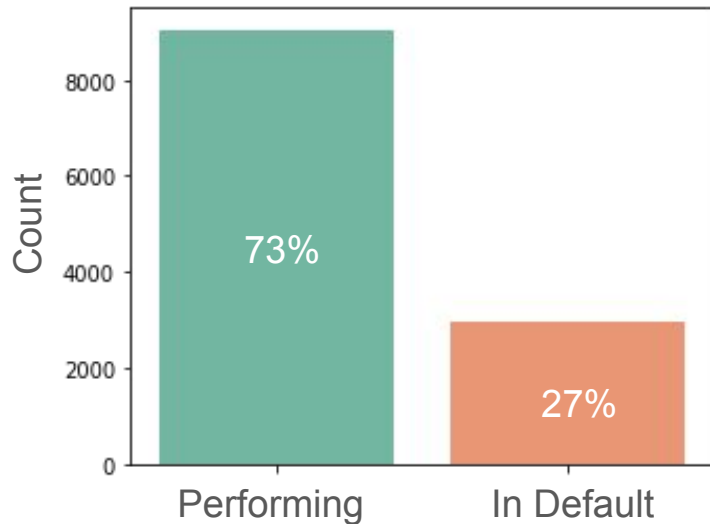
Our Modeling Approach

Modeling Workflow Entertained a Wide Variety of Models with Targeted Tuning



Three Main Constraints Related to Data Imbalance were mitigated

Fig 4: Distribution of Defaults
Versus Performing/Paid Off



Challenge 1: Imbalanced Data

- We did not use SMOTE → overfitting
- Instead, we used **class-weight rebalancing**

Challenge 2: Balancing Risk vs. Growth

- False Negatives created direct financial loss → Recall

Challenge 3: Lowering Threshold to Boost Recall

- Results is a drop in precision, so more good customers are rejected

Our Solution: Threshold Optimization

- Evaluate thresholds along the **Precision-Recall Curve**
- Select the best **balance** using the **F1 Score**

Logistic Regression Preferred Upon Model Comparison

Model	Threshold	ROC-AUC	Accuracy	F1-Score	Recall	Precision
Logistic Regression	0.4684	0.5901	0.4608	0.3412	0.7264	0.2229
Random Forest	0.4699	0.5838	0.4898	0.3350	0.6685	0.2235
XGBoost	0.4792	0.5830	0.5047	0.3364	0.6530	0.2265

Fig 5:
Model Metrics

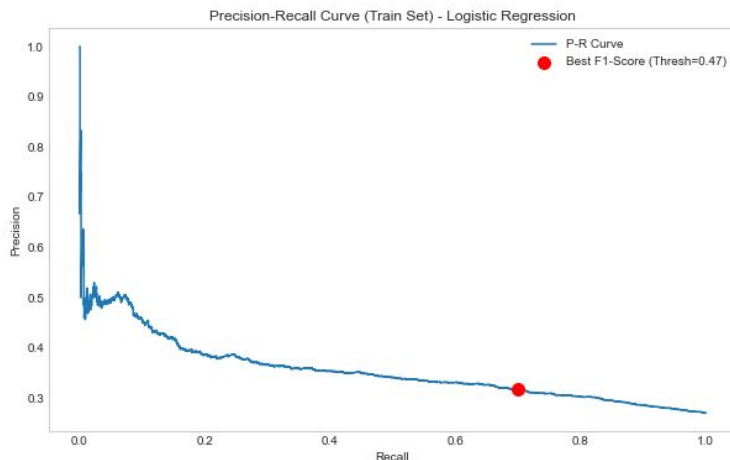


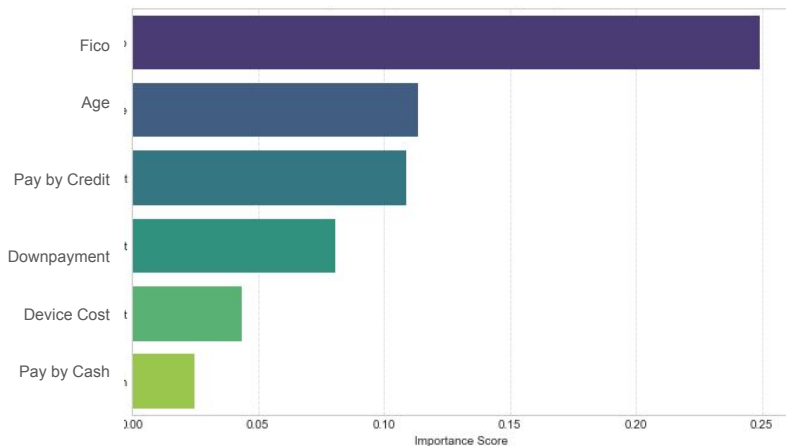
Fig 6: Precision-Recall Curve - Logistic Regression

- Best Overall Performance:**
 It delivered the strongest results: **ROC-AUC (59.01%)**, **F1-Score (34.12%)**, and **Recall (72.64%)**.
- Critical Interpretability:**
 It is a transparent "White Box" model, allowing us to explain *why* a decision was made, unlike "Black Box" models .
- Actionable Customer Feedback:**
 This transparency lets us provide constructive feedback (e.g., "Improve FICO score") to rejected applicants, reducing customer frustration.
- No Missing Data:**
 The dataset contains no missing value and shows stable, well-distributed figures.

Model Interpretation

Our Model Presents a Clear Hierarchy of Predictors

Fig 7: Feature Importance Scores



Feature	Coefficient	Interpretation
Intercept	-0.0302	Baseline default probability when all features are 0.
FICO Score	-0.2493	Higher FICO score strongly reduces default risk.
Borrower Age	-0.1136	Older borrowers are less likely to default .
Down Payment	-0.0809	Customers who make a larger upfront payment show stronger commitment and have lower default risk .
Pay by Credit Card	-0.1087	Paying via credit card is associated with lower risk .
Pay by Cash	+0.0247	Weak positive association with default risk.
Device Cost	+0.0435	More expensive devices slightly increase default risk.

Fig 8: Feature Coefficient Interpretation

Our Model Balances Predicting Default with Predicting Repayment

Fig 8: Confusion Matrix for Final Model

	Predicted Performing	Predicted Default
Actual Performing	1185	1795
Actual Default	194	515

- **1185** customers who ***will pay*** are correctly approved → supports business growth.
- **515** customers will default The model correctly flags many high-risk customers → meaningful risk reduction.
- The model **misses 194 defaulters** → relatively little financial loss.
- **1795** good customers are rejected → the trade-off between growth and risk.



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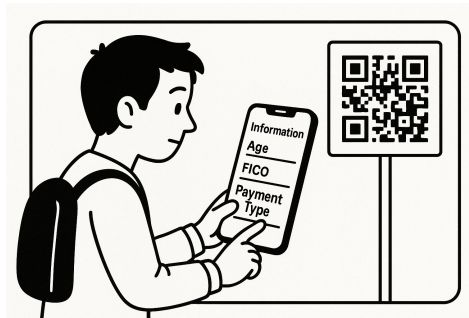
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Our Business Case: What This Means in Practice

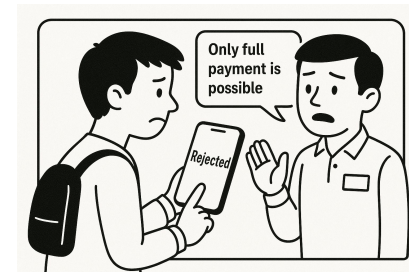
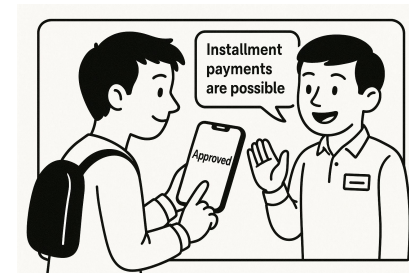
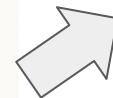
Verizon Approval/Rejection Process



Select product



Provide and verify information



Get result from model

Business Benefits:

1. **Accuracy:** Fewer wrong approvals or rejections.
2. **Efficiency:** Faster, automated decision-making.
3. **Revenue:** Better targeting boosts conversions.
4. **Satisfaction:** Quick and fair approvals enhance experience.

Business Costs:

1. **Setup:** Data preparation, model training
2. **Maintenance:** Continuously updating thresholds and the model based on new customer data

The Business Value of Our Model

Assuming

- Customer LTV = $0.1 * \text{device_cost} = \94.5
- Cost of Default = $(24-9) / 24 * \text{device_cost} + \text{service_cost} = \740.6
 - Total loan term: 24 months
 - Default Timing: After 9 months

$$\Delta \text{ Value} = \Delta \text{ Profit} - \Delta \text{ Loss}$$

$$\begin{aligned} &= (\text{TN}(\text{true negatives}) \times \text{Customer LTV}) \\ &+ (\text{TP}(\text{true positives}) \times \text{Cost of Default}) \\ &- (\text{FN}(\text{false negatives}) \times \text{Cost of Default}) \\ &- (\text{FP}(\text{false positives}) \times \text{Customer LTV}) \end{aligned}$$

TOTAL Value = \$180,096
(Per customer without scaling)

Business Impact of Our Model in Terms of Increased Cash Flow

Based on a hypothetical base of 1M new customers

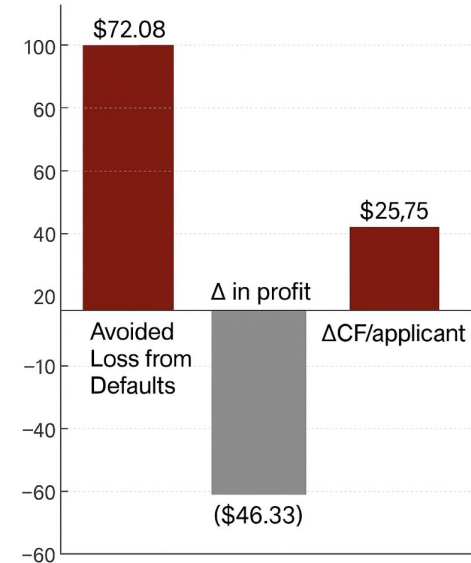
$$\Delta \text{ Cash Flow} = \text{TN} \times \text{LTV} - \text{FN} \times \text{Cost of default}$$

	Current model	Our Model	Net Change
Total Customers (TN+FN)	780,000	373,800	(406,200)
# of Paying Customers (TN)	630,089	321,213	(308,876)
Profit from Paying customers	\$94,513,418	\$48,181,980	(\$46,331,439)
Default Customers (FN)	149,910.54	52,586.80	(97,323.74)
Loss from defaults	(\$111,023,750)	(\$38,945,786)	\$72,077,964
Total Cash Flow	(\$16,510,331)	\$9,236,194	\$25,746,525
Cash Flow per applicant	(\$16.51)	\$9.24	\$25.75

Fig 9: Net Cash Flow Per 1 million applicant

This shows a \$25.75 increase per applicant and decrease in profit offset by a huge minimization in default costs

Fig 10: Δ Cash Flow Per Applicant



Requests, Summary, & Timeline



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Business Recommendations Based on the Model

- Re-optimize the approval threshold using selected model on a quarterly basis
- After 8 payments, at-risk customers are sent a “check-in” email
- Applicants under 20 need a cosigner with good credit
- Incentivize high-Fico applicants with free upgrades at more regular intervals than average customers



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Further Requests for Data to Improve Model Accuracy

- Exact number of months each customer has been with the company?
- Number of active lines on an account?
- How often does a customer upgrade their device?

Nov 20

We update our model to avoid overlap and conflicts with the current algorithm

Dec 5

We update our model based on the newly provided variables

Nov 15

Client provides more info on their current application algorithm

Nov 30

Client provides more info on follow up questions

Dec 15

Final review and meeting



Appendix

Further data requests that could benefit model accuracy (description)

The question and answer report mentions that existing customers have an average tenure of 4.2 years, but an average alone doesn't tell the full story. If Verizon provides the exact number of months each customer has been with the company, the model can learn more realistic patterns. for example, that customers become much less likely to default after about two years, but the benefit of tenure levels off once they've been with Verizon for five years or more.

Customers who have several active lines or bundled Verizon services, like internet, TV, or family plans, tend to be more stable and less likely to default. Having the number of active lines on an account would help the model identify loyal, deeply engaged customers who are generally safer to finance.

How often a customer upgrades their device can reveal a lot about their stability. Someone who upgrades every year shows consistent engagement and tends to stay current on payments, while someone who upgrades only once every five or six years may behave very differently. Having upgrade timing or frequency would help the model capture these patterns.